

Business Problem Statement

Healthcare providers face a significant challenge with **patient no-shows**, where patients fail to attend scheduled appointments without prior cancellation. These missed appointments lead to underutilized medical resources, increased operational costs, longer waiting times for other patients, reduced revenue, and disruptions in healthcare service delivery.

Patient no-shows are a major operational challenge for healthcare providers. When patients miss scheduled appointments without prior notice, clinics experience wasted appointment slots, inefficient use of medical staff, increased operational costs, and longer waiting times for other patients seeking care.

The healthcare organization wants to analyze appointment records to understand the factors associated with patient attendance and no-show behavior. By examining patient demographics, appointment scheduling patterns, attendance history, accessibility factors, weather conditions, and satisfaction levels, the organization aims to identify trends and insights that can help improve appointment management and reduce missed visits.

Business Objectives

- Understand the overall no-show rate across appointments.
- Identify the key factors associated with missed appointments.
- Analyze how waiting time impacts attendance behavior.
- Examine the influence of previous no-shows on future attendance.
- Evaluate the effect of SMS reminders on appointment attendance.
- Investigate whether travel distance and weather conditions contribute to no-show rates.
- Assess the relationship between patient satisfaction and attendance.
- Provide actionable recommendations to improve appointment attendance and clinic efficiency.

Key Business Questions

1. What percentage of appointments result in a no-show?
2. Which patient age groups have the highest no-show rates?
3. Does gender influence appointment attendance?
4. How does waiting time affect the likelihood of missing an appointment?
5. Are patients with a history of no-shows more likely to miss future appointments?
6. Do SMS reminders reduce no-show rates?
7. Does travel distance impact appointment attendance?
8. How do weather conditions affect patient attendance?
9. Is there a relationship between patient satisfaction and no-show behavior?
10. Which factors appear to have the strongest association with missed appointments?

Expected Outcome

Provide data-driven insights into patient attendance behavior and identify the factors most associated with no-shows, enabling healthcare administrators to improve scheduling processes, patient communication strategies, and overall clinic resource utilization.